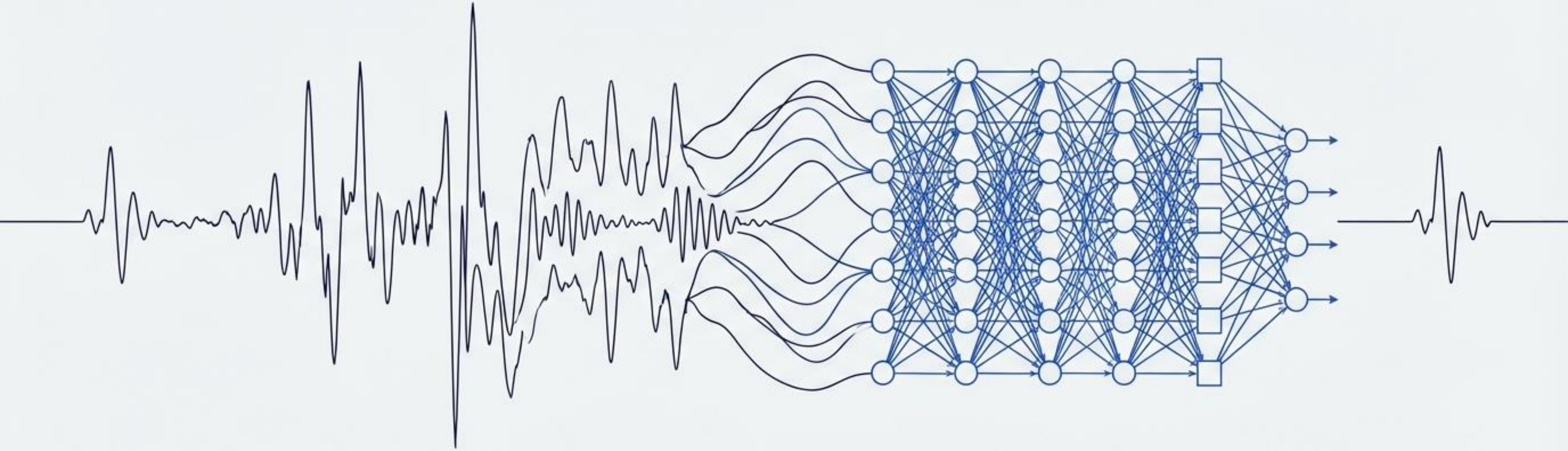

Deep Learning for the Detection of Seizures in EEG Signals

A Comparative Analysis of Rule-Based, Machine Learning, and Neural Network Approaches

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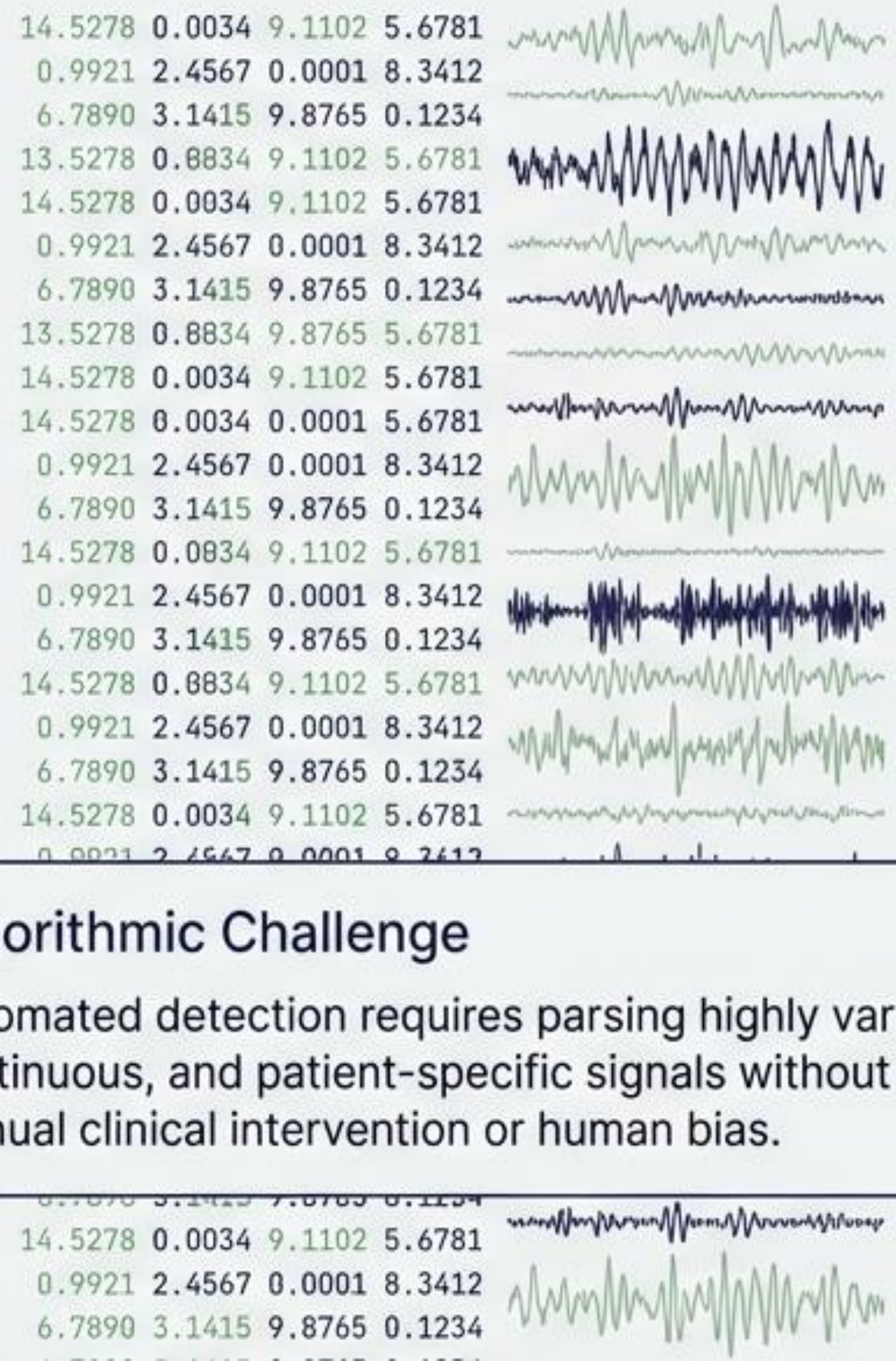
Institution: Universitat Autònoma de Barcelona (UAB)

Course: Deep Learning

The Challenge of Automating Detection



Clinical Context

Epileptic seizures represent brief, abnormal electrical activity in the brain. They are typically diagnosed through manual review of EEG recordings.

Algorithmic Challenge

Automated detection requires parsing highly variable, continuous, and patient-specific signals without manual clinical intervention or human bias.

Finding the Needle in a 570,000-Window Haystack

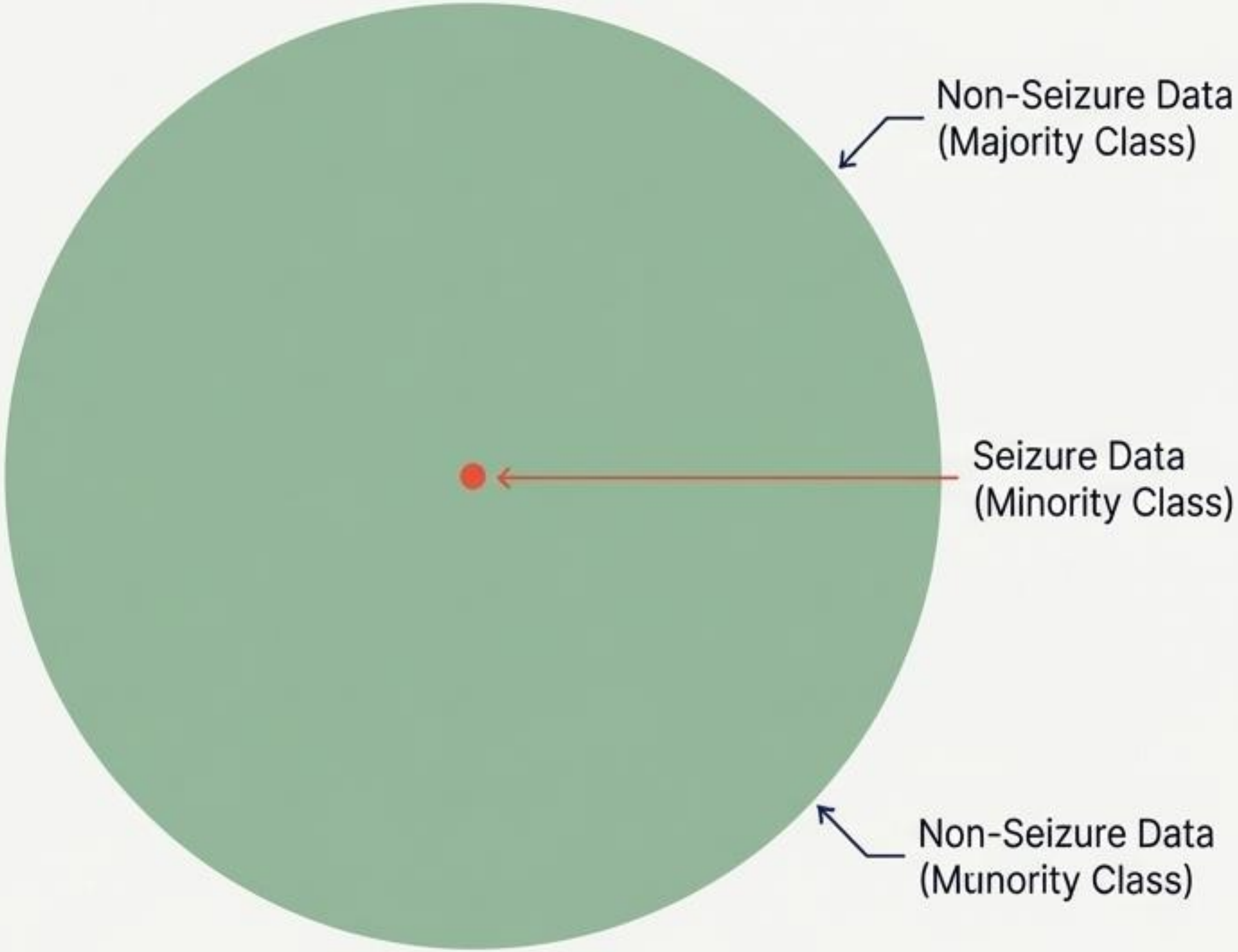
CHB-MIT Dataset Anatomy

- **Source:** Scalp EEG database from 24 patients
- **Segmentation:** Non-overlapping 1-second windows

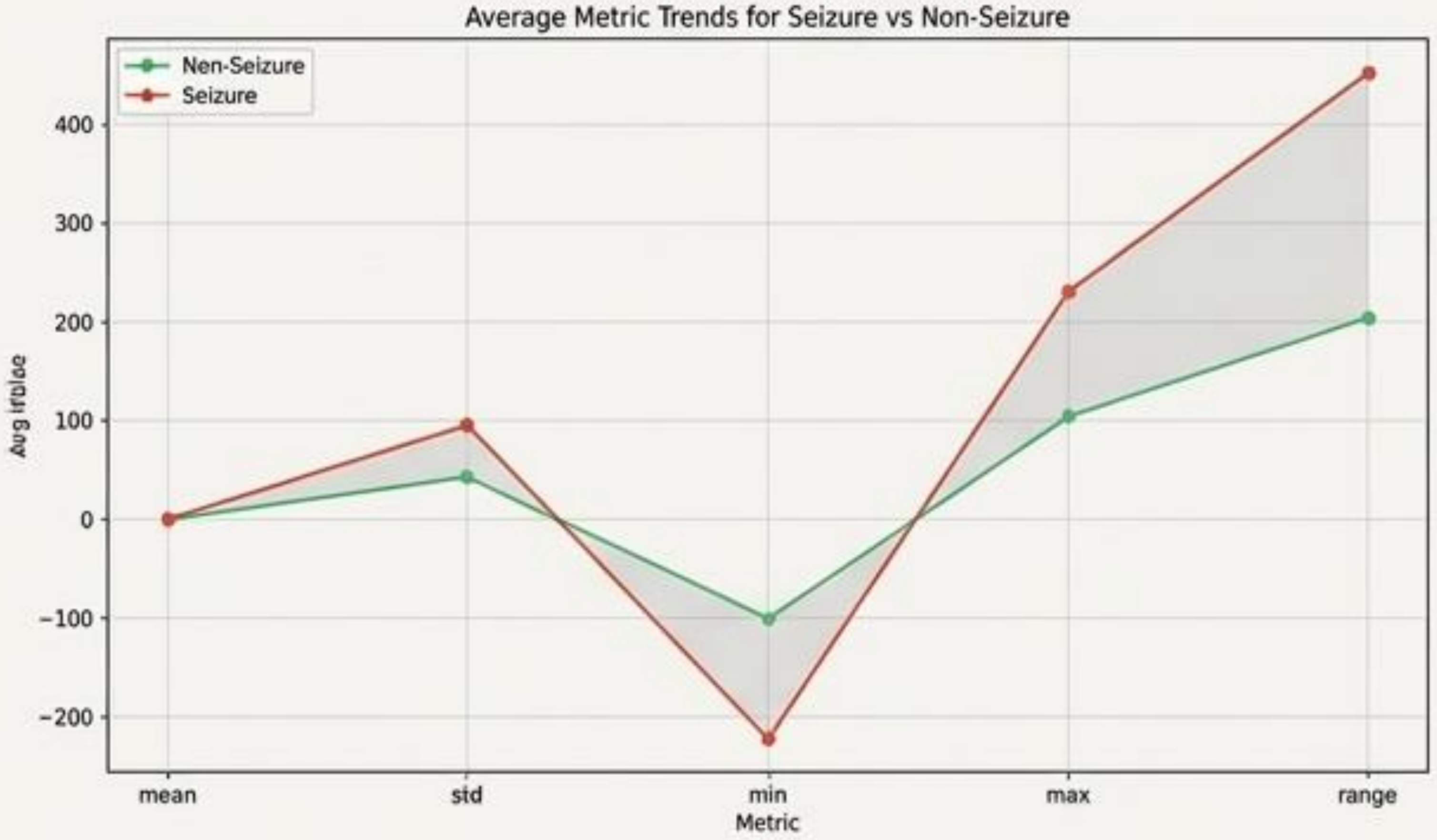
568,425
Non-Seizure Samples

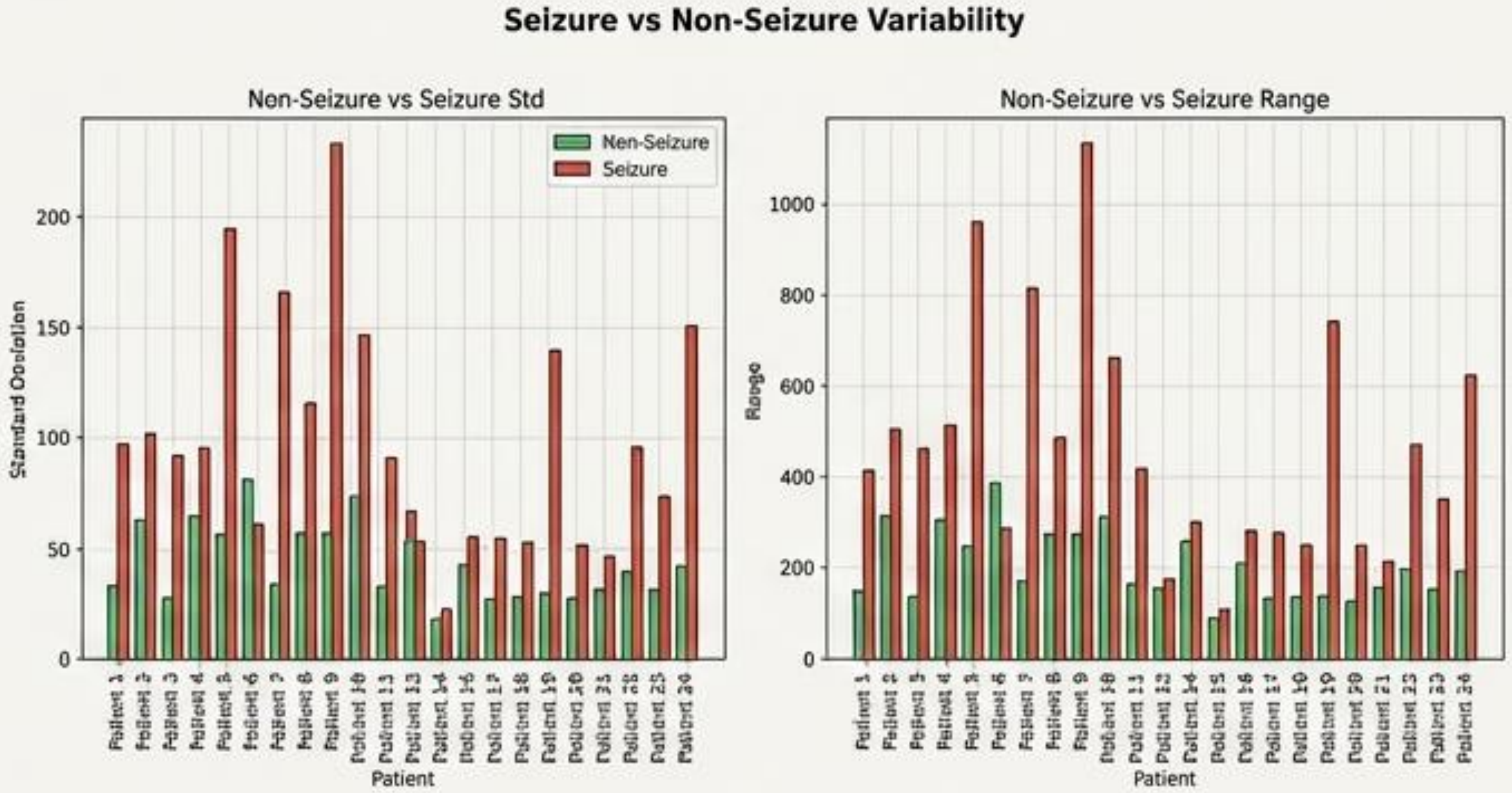
3,480
Seizure Samples (0.6%)

The Imbalance Problem: Models must detect an incredibly rare anomaly without being overwhelmed by the massive volume of normal brain activity.



The Statistical Signature of a Seizure is Extreme Dispersion





Exploratory Insight

Data analysis reveals that seizure windows exhibit drastically higher standard deviation and range compared to non-seizure windows across nearly all 24 patients.

Algorithmic Implication

Because the signal physically scatters, all three evaluated models must be structurally sensitive to sudden, extreme spikes in signal variability.

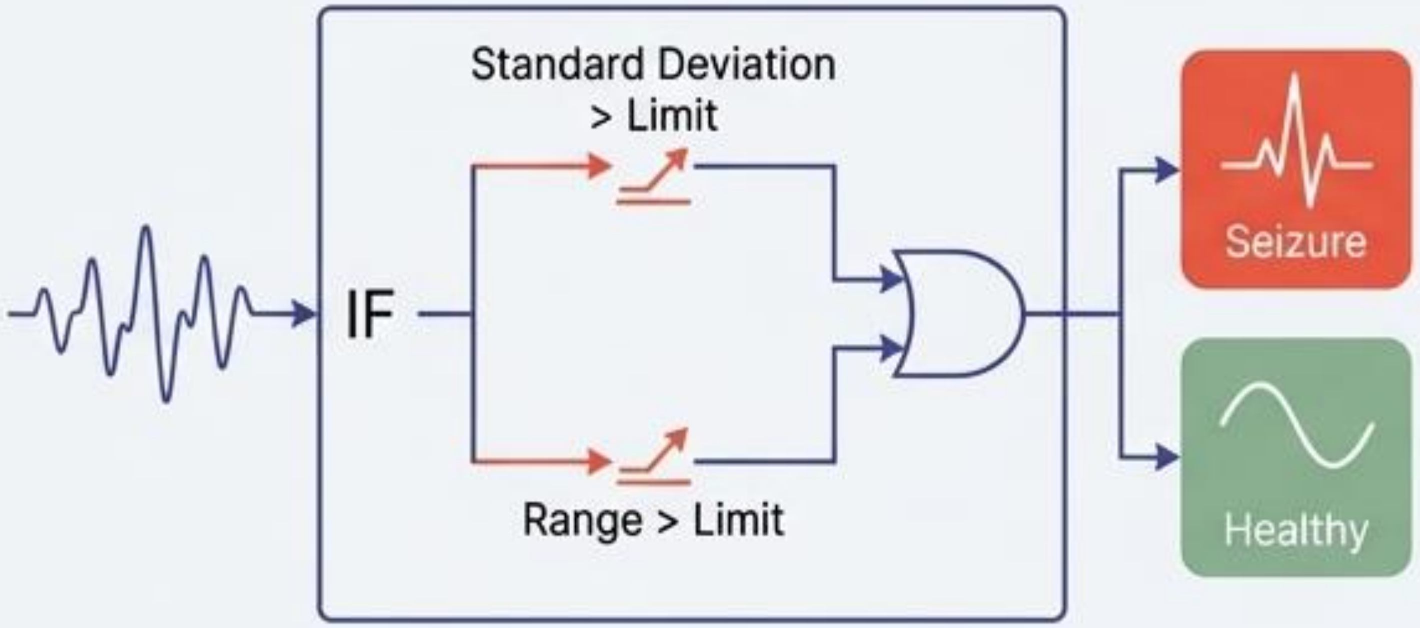
Three Approaches to Analyzing EEG Windows

	Statistical Rules (Threshold)	**Traditional ML (Random Forest)**	**End-to-End DL (1D CNN)**
Input	CHB-MIT EEG windows `EEGDataset` (normalize=False) per patient	CHB-MIT EEG windows `EEGDataset` (normalize=False) per patient	CHB-MIT EEG windows `EEGDataset` (normalize=False) per patient
Preprocessing / Features	`extract_basic_features` → mean/std/min/max/range `compute_thresholds` per fold (std/range/min/max averages)	`extract_basic_features` → mean/std/min/max/range, `peak_to_peak`, `std_range_ratio`, `range_std_sum` Feature vector ordering preserved	Window normalization (zero mean, unit std) per 1s segment `DataLoader` (batch=32, shuffle, pin_memory)
Classifier	`ThresholdClassifier` rule-based (std ≥ thr, range ≥ thr, optional min/max bounds) Patient-wise k-fold evaluation	`RandomForestClassifier` (`n_estimators=100`, `class_weight='balanced'`, `depth=None`) Patient-wise k-fold evaluation & accuracy metrics	`SimpleEEGCNN` (Conv1d×3 → BatchNorm → ReLU → MaxPool) FC layers → Dropout(0.3) → Softmax Adam(lr=1e-3) + CrossEntropyLoss with class weights Save fold checkpoint → out/models/cnn Patient-wise k-fold eval (loss, accuracy per fold)

CNN bypasses manual feature extraction entirely.

Collapsing Time into Static Numbers

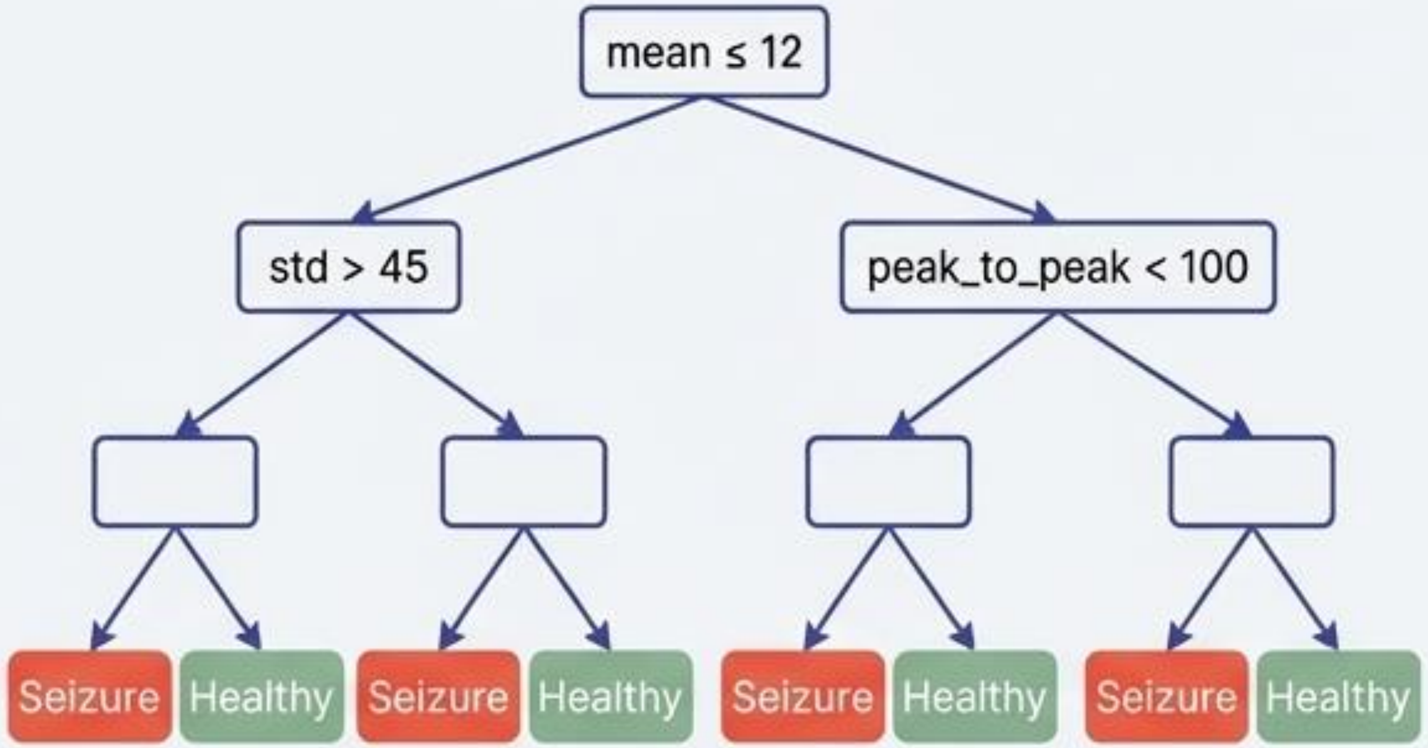
Approach 1: Threshold



Mechanism: Flags a 1-second window if its standard deviation or range exceeds fold-specific optimized limits.

Drawback: Extremely rigid. It only looks at the mathematical limits, ignoring the shape of the wave.

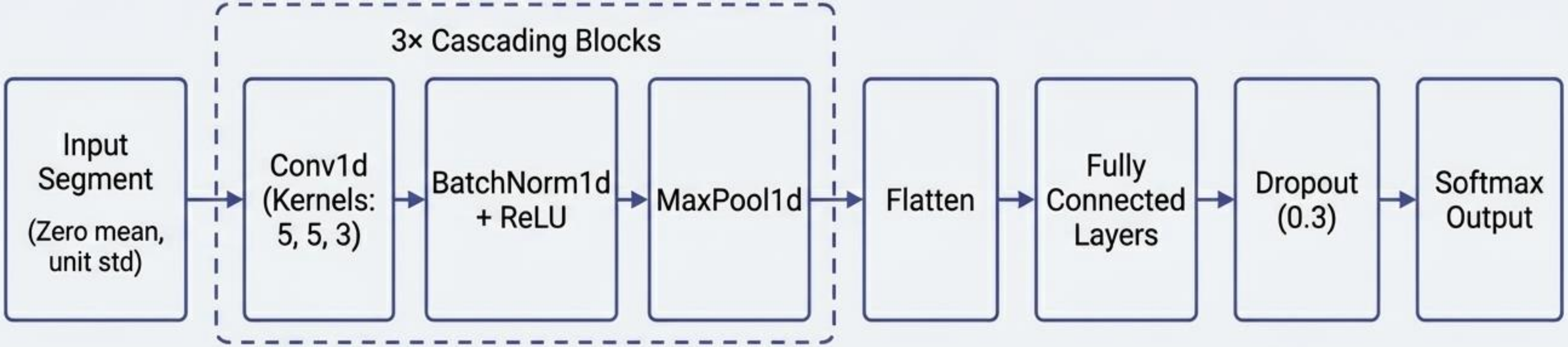
Approach 2: Random Forest



Mechanism: Computes 100 estimators operating on handcrafted time-domain features.

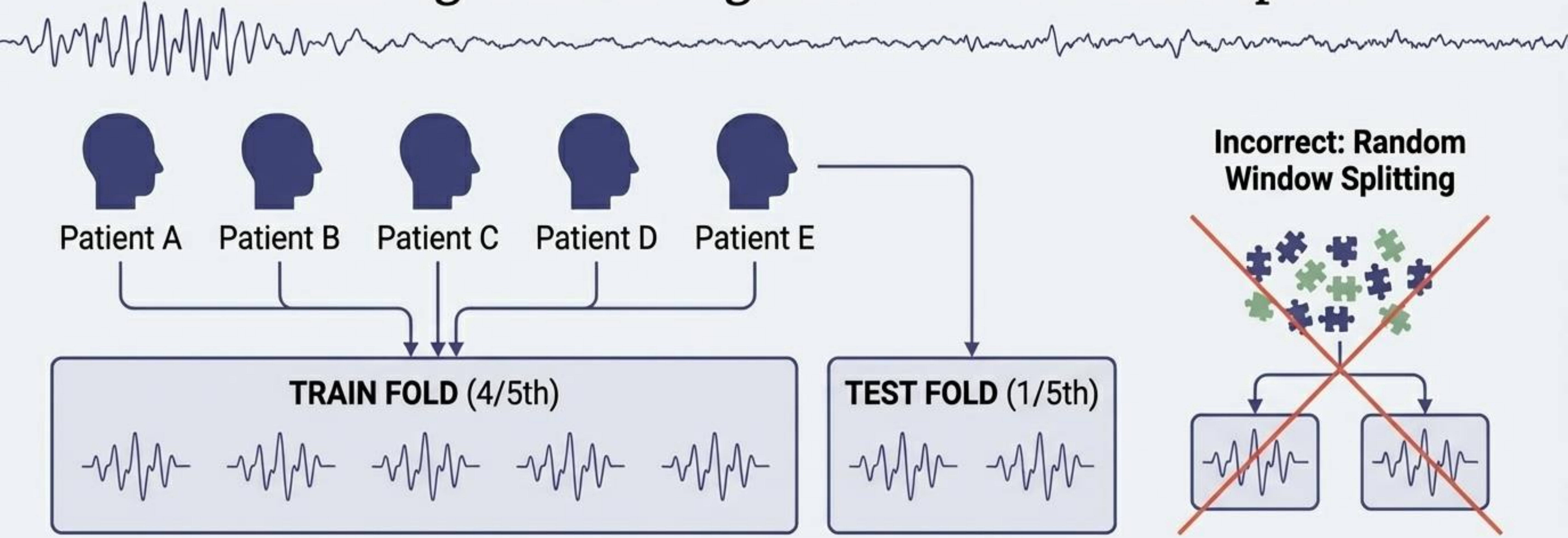
Drawback: Smarter classification, but effectively destroys the sequential time-domain nature of the signal by flattening 1 second of data into a few basic averages.

End-to-End Deep Learning Learns the Sequence



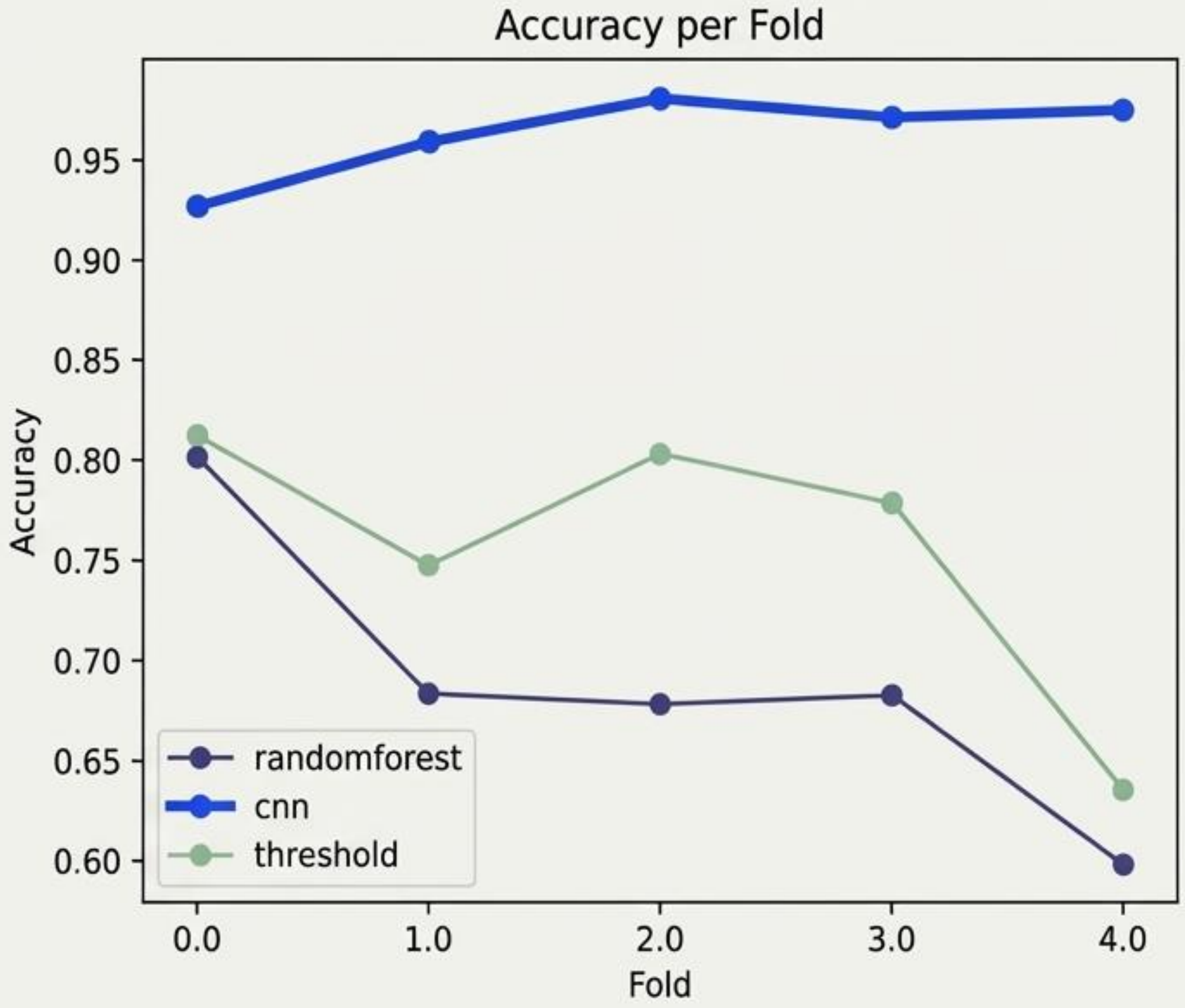
Mechanism	Optimization	Advantage
Normalizes the raw 1-second segment and feeds it directly into a One-Dimensional CNN.	Trained with Adam optimizer (learning rate 10^{-3}) for 50 epochs.	Completely bypasses human feature engineering, allowing the network to act as a learnable filter for temporal patterns.

Preventing Data Leakage with Patient-Level Splits



- **Method:** 5-Fold Patient-Level Cross Validation.
- **The Rule:** Windows from each specific patient appear in exactly one test fold.
- **The Reason:** Randomly splitting windows mixes a single patient's data across train and test sets, artificially inflating accuracy. Patient-level CV ensures the model generalizes to entirely unseen brains.

Deep Learning Remains Stable Across Unseen Patients



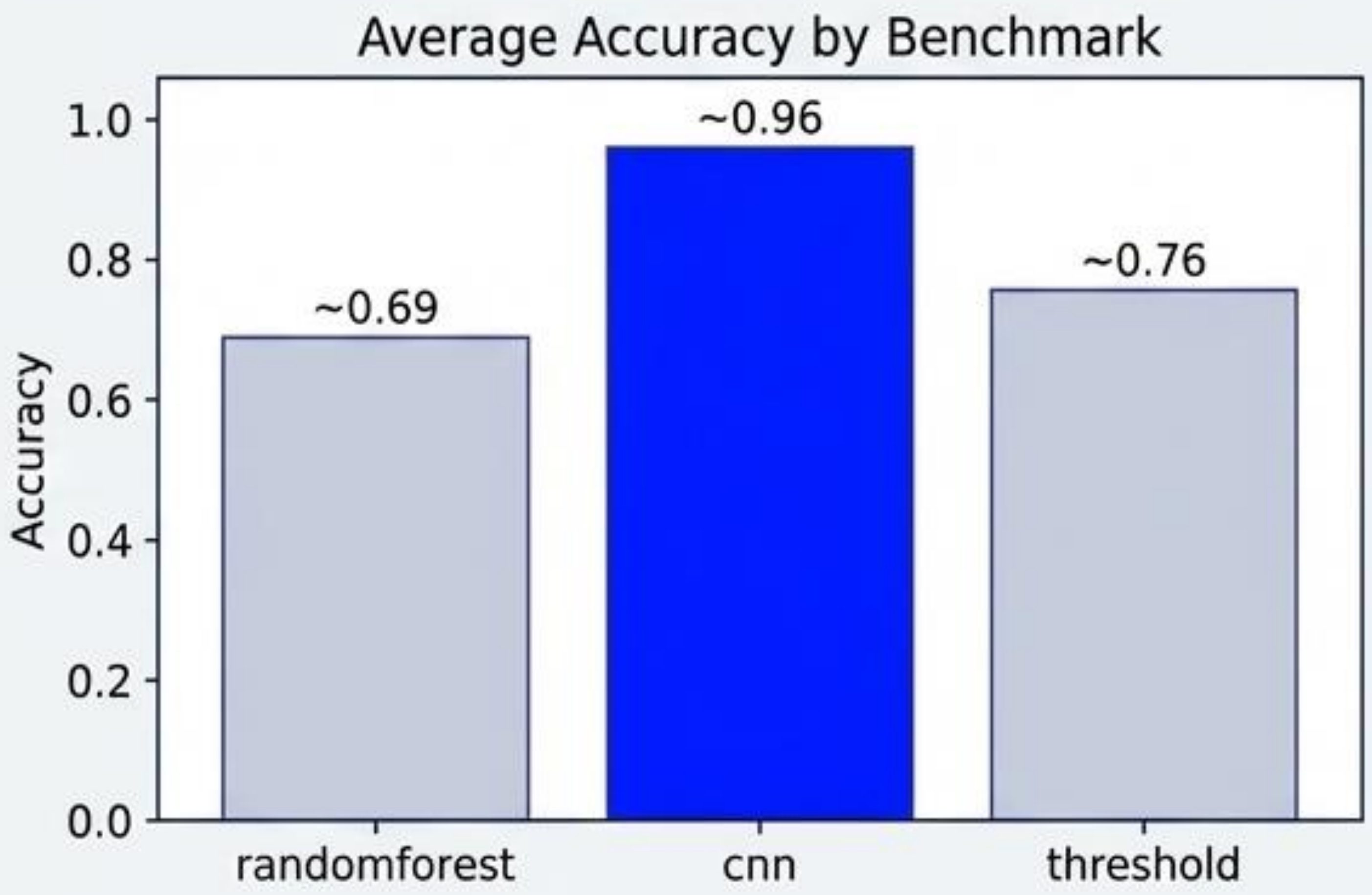
Observation:

The CNN demonstrates remarkable stability across all 5 patient hold-out sets, consistently maintaining an accuracy above 0.90.

Contrast:

The Random Forest suffers from massive accuracy swings. It fails to generalize consistently when faced with the unique, unseen EEG patterns of new patients in different folds.

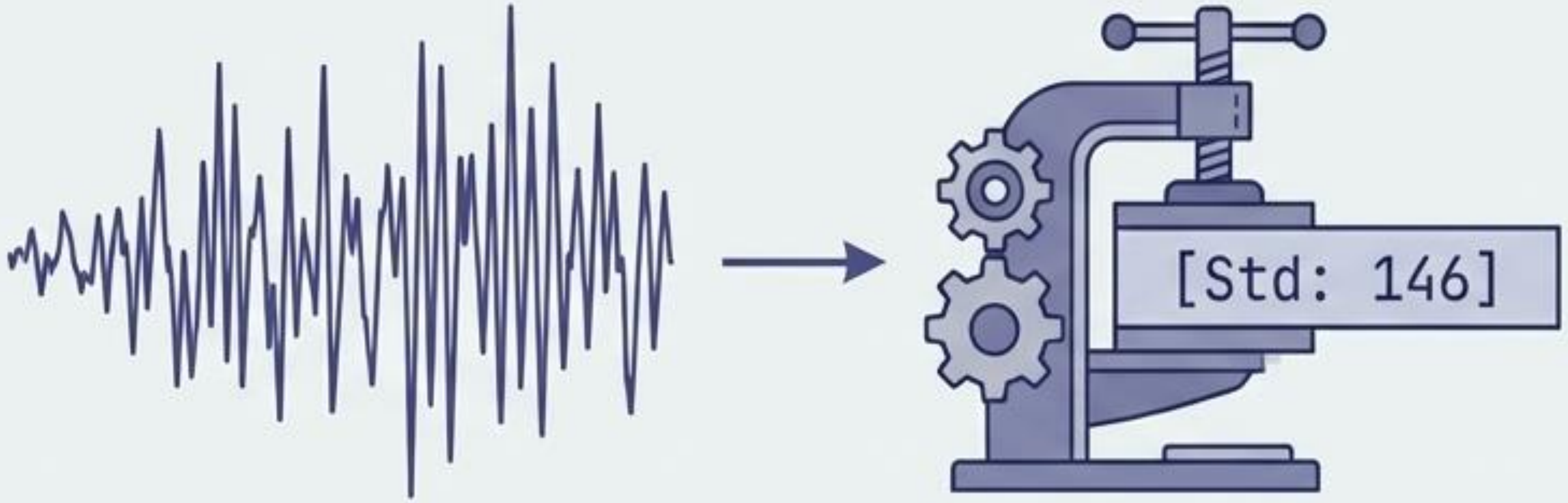
The CNN Conclusively Outperforms Alternative Baselines



The Verdict

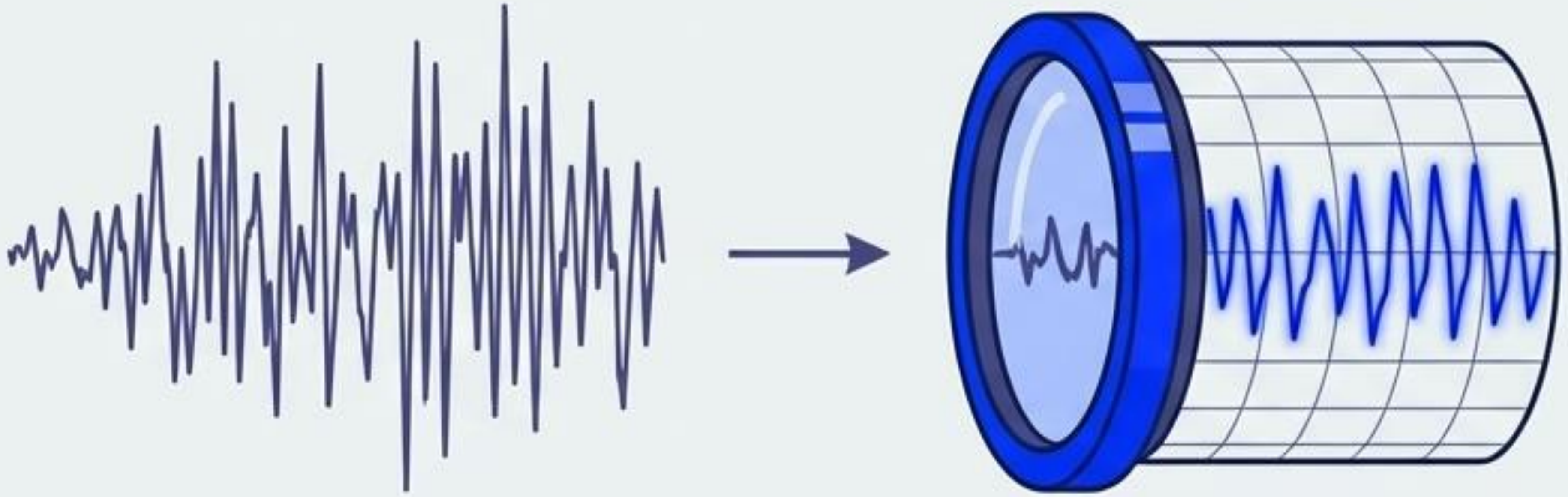
The deep learning model leads the pack by a massive margin in average accuracy across folds. This definitively proves the value of expressive, non-linear representation learning over handcrafted feature engineering for continuous EEG analysis.

Why End-to-End Models Win



The Flaw of Handcrafted Features

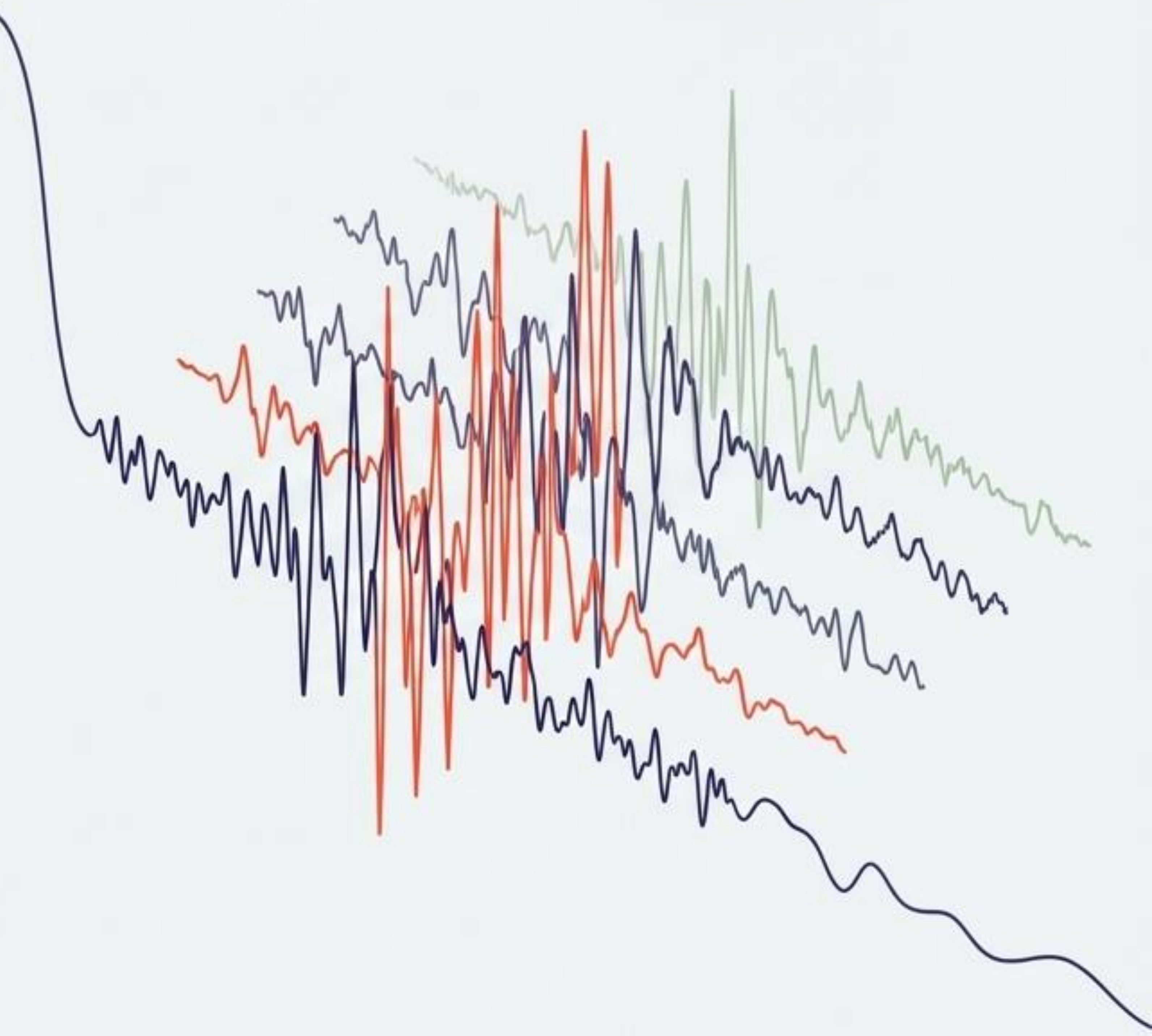
Collapsing a 1-second electrical sequence into basic averages (Random Forest / Threshold) destroys vital temporal patterns and sequential context.



The Power of Convolutions

1D convolutions act as learnable spatial filters. They do not crush the data; they slide across it, detecting the specific sequential rhythmic signatures of seizures over time.

Conclusion & Future Horizons



Summary

Expressive, end-to-end convolutional neural networks provide superior automated seizure detection compared to traditional handcrafted machine learning.

Future Horizon I: Spatial Data

Integrating multichannel spatial information to track electrical signal propagation across different regions of the scalp.

Future Horizon II: Temporal Context

Expanding the temporal context beyond isolated 1-second windows to understand broader seizure onset dynamics and pre-ictal states.